Supplementary Figures

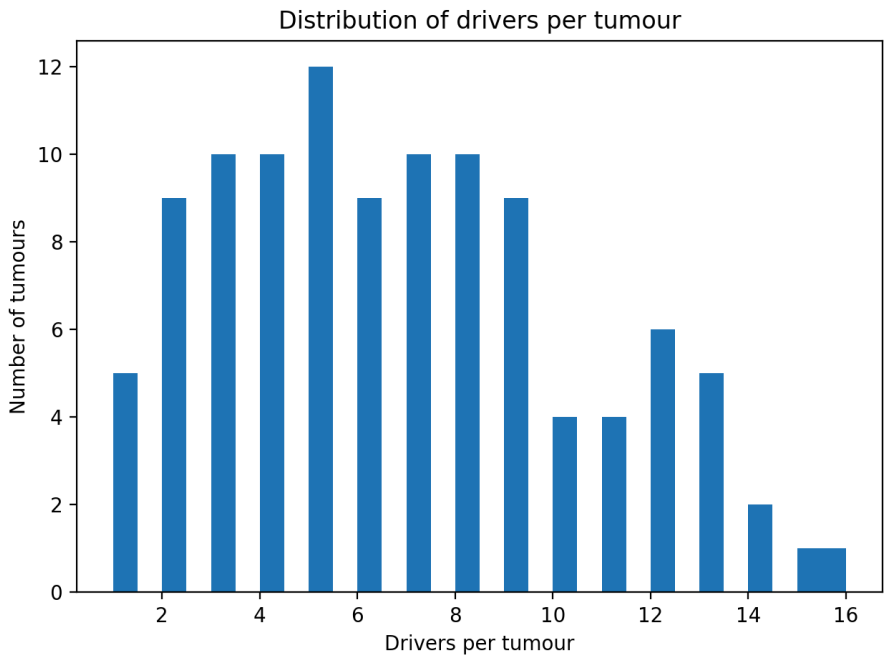

Figure 1. Empirical distribution of the number of driver genes per tumour derived from PCAWG/ICGC breast adenocarcinoma samples.

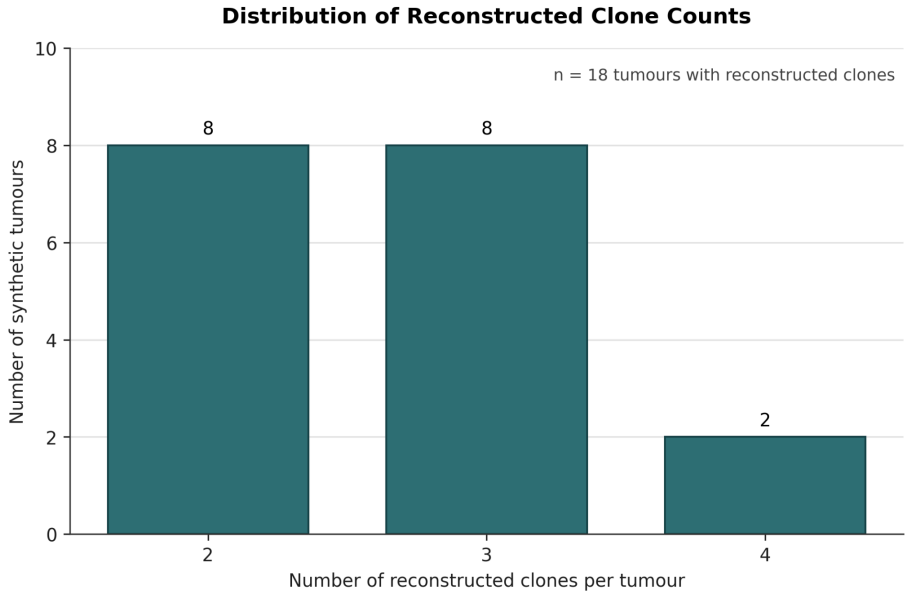

Figure 2. Distribution of the number of reconstructed clones per synthetic tumour.

Supplementary Tables

Supplementary Table 1. Clinical Characteristics of one representative synthetic breast cancer cohort generated by the pipeline.

Characteristic	n	&	Notes
Total breast cancer patients	39	100	Patients with malignant neoplasm of breast diagnosis
Breast biopsy	20	51.3	Patients had biopsy
Stage I	7	17.9	
Stage II	3	15.4	
Stage III	4	10.3	
Stage IV	3	7.7	
HER2-positive	6	15.4	Positive IHC or FISH greater than 2.2
ER-positive	16	41.0	Positive estrogen receptor IHC
PR-positive	12	30.8	Positive progesterone receptor IHC
Triple-negative phenotype	2	5.1	ER-negative, PR-negative, and HER2-negative
Latest treatment response: Improving	21	53.8	Latest recorded response per patient
Latest treatment response: Worsening	4	10.3	Latest recorded response per patient
Breast surgery	14	35.9	Lumpectomy or mastectomy
Chemotherapy	13	33.3	Chemotherapy medication or chemotherapy

			procedure exposure
Radiation therapy	18	46.2	Radiation therapy or brachytherapy procedure
Endocrine therapy	17	43.6	Tamoxifen, aromatase inhibitor, or fulvestrant
HER2-targeted therapy	7	17.9	Trastuzumab, lapatinib, or neratinib
CDK4/6 inhibitor therapy	11	28.2	Palbociclib, ribociclib, or abemaciclib exposure
Two or more sequencing timepoints	18	46.2	
Three sequencing timepoints	5	12.8	
Reconstructed clone groups	18	46.2	
Breast-cancer cause of death	6	15.4	

Supplementary Table 2. Synthea drugs gene resistance/sensitive associations, used in the generator, along with references.

Drug(s)	Association	Gene	Mutation type	References
Epirubicin	Resistance	TP53, CHEK2	Disruptive/inactivating TP53; loss-of-function CHEK2	TP53, CHEK2
Doxorubicin	Resistance	TP53, CHEK2	Disruptive/inactivating TP53; somatic nonsense	TP53, CHEK2

Tamoxifen + Aromatase Inhibitors (Letrozole, Anastrozole, Exemestane)	Resistance	NF1	Loss of function	Reference
Endocrine Therapies	Resistance	ESR1	Fusion	Reference
Endocrine therapies	Resistance	RUNX1, CTCF, ARID1A	Gene disruptive	Reference
Fulverstrant	Resistance	ARID1	Loss of Function	Reference
Aromatise Inhibitors (Letrozole, Anastrozole, Exemestane)	Sensitivity	GATA3, MAP3K1, MAP2K4	Somatic coding mutations broadly	Reference
Aromatise Inhibitors (Letrozole, Anastrozole, Exemestane)	Resistance	TP53	Functionally damaging TP53 alterations	Reference
Fulvestrant	Resistance	ARID1A	Inactivating	Reference
Trastuzumab	Resistance	PTEN	Loss of Function	Reference
Trastuzumab	Resistance	PIK3CA	Activating mutation	Reference
Lapatinib	Resistance	PTEN	Loss of Function	Reference
Lapatinib	Resistance	PIK3CA	Inactivating	Reference
Palbociclib + Ribociclib + Abemacicilib	Resistance	RB1, FAT1	Loss of Function	Reference
Palbociclib + Ribociclib + Abemacicilib	Resistance	PTEN, NF1	Pathway escape	Reference
Doxorubicin	Sensitivity	TP53	Missense Mutation and Protein Altering Variant	CIViC, CIViC
Fulvestrant	Sensitivity	PIK3CA	Various	CIViC, CIViC, CIViC,

				CIViC, CIViC, CIViC, CIViC, CIViC, CIViC, CIViC, CIViC, CIViC, CIViC, CIViC, CIViC, CIViC, CIViC, CIViC, CIViC
Fulverstrant	Sensitivity	AKT1	Missense Mutation	CIViC
Fulverstrant	Sensitivity	ESR1	Missense Mutation	CIViC, CIViC, CIViC, CIViC
Trastuzumab	Sensitivity	PIK3CA	Missense Mutation	CIViC
Trastuzumab	Sensitivity	FCGR3A	Missense Mutation	CIViC, CIViC
Trastuzumab	Sensitivity	ERBB3	Missense Mutation	CIViC
Lapatinib	Sensitivity	PIK3CA	Missense Mutation	CIViC
Lapatinib	Sensitivity	ERBB3	Missense Mutation	CIViC
Neratinib	Sensitivity	ERBB2 (HER2)	Various	CIViC, CIViC,CIViC, CIViC
Palbociclib	Sensitivity	PIK3CA	Various	CIViC, CIViC, CIViC, CIViC, CIViC, CIViC, CIViC, CIViC
Palbociclib	Sensitivity	ESR1	Various	CIViC, CIViC, CIViC, CIViC, CIViC, CIViC, CIViC, CIViC, CIViC, CIViC
Tamoxifen	Sensitivity	ESR1	Various	CIViC, CIViC, CIViC, CIViC, CIViC
Letrozole	Sensitivity	ESR1	Missense Mutation	CIViC
Exemestane	Sensitivity	STK11	Missense Mutation	CIViC
Ribociclib	Sensitivity	PIK3CA	Transcript Variant, Gain Of Function Variant	CIViC
Abemaciclib	Sensitivity	ESR1	Not provided	CIViC
Tamoxifen	Resistance	TP53	Missense Mutation	CIViC
Trastuzumab	Resistance	PIK3CA	Missense Mutation	CIViC, CIViC, CIViC, CIViC, CIViC, CIViC,

				CIViC, CIViC, CIViC, CIViC, CIViC, CIViC, CIViC, CIViC, CIViC
Trastuzumab	Resistance	ERBB2 (HER2)	Not specified	CIViC, CIViC, CIViC
Lapatinib	Resistance	ERBB2	Various	CIViC, CIViC, CIViC
Lapatinib	Resistance	PIK3CA	Missense mutation	CIViC, CIViC, CIViC, CIViC, CIViC
Neratinib	Resistance	ERBB2	Missense mutation	CIViC
Palbociclib	Resistance	RB1	Frameshift Truncation	CIViC
Ribociclib	Resistance	RB1	Not specified	CIViC

Supplementary Methods

2.1 Synthetic clinical data generation

The clinical component of the dataset was generated by the open-source patient simulator Synthea. Synthea produces fully synthetic patient output using only publicly available information and health statistics as input. Instead of focusing on one problem or disease, it simulates the longitudinal healthcare history of each patient, including encounters, diagnoses, observations, procedures, medications, care plans, immunizations and more. The resulting data are realistic and free of privacy and security restrictions (Walonoski *et al.* 2018).

For this work, Synthea version 3.3.0 was used. To restrict the cohort to breast cancer, Synthea was run using its disease-specific filtering option so that only patients with breast cancer were retained. Although Synthea can export data in FHIR format, this study used the CSV output because it provides relational tabular files that are easier to process and link with the synthetic genomic component. The CSV export contains multiple files connected by patient identifiers rather than a single record per patient.

The Synthea breast cancer module provides several clinically useful variables for this project, including cancer stage, hormone receptor status, HER2 status, surgical procedures, chemotherapy, radiation therapy, hormone therapy, HER2-targeted therapy, CDK4/6 inhibitor

therapy and follow-up events. The module also includes a treatment-response observation, recorded as “Improving” or “Worsening”. This response variable was used as one of the anchors for linking the clinical trajectory with the synthetic tumour genomic profile.

2.2 Breast cancer driver gene set

To extend Synthea’s clinical cohort with tumour genomics, a set of breast cancer driver genes was defined. The goal was not to assign mutations independently of the clinical data, but to make the genomic profile partly dependent on the patient’s simulated tumour phenotype, treatment context and treatment-response trajectory.

An empirical distribution of the number of driver genes per tumour was derived from the Pan-Cancer Analysis of Whole Genomes/International Cancer Genome Consortium (PCAWG/ICGC) panorama of driver mutations file “TableS3_panorama_driver_mutations_ICGC_samples.public.tsv.gz”, downloaded from the ICGC 25K open-access object bucket (ICGC ARGO, n.d.). The file reports high-confidence driver events identified across PCAWG tumours (Zhang *et al.* 2019) (The ICGC/TCGA Pan-Cancer Analysis of Whole Genomes Consortium *et al.* 2020). For this step, only breast adenocarcinoma samples were used. The resulting distribution was used to sample the total number of driver genes assigned to each synthetic patient, with a maximum of 16 driver genes per tumour (Supplementary Figure 1).

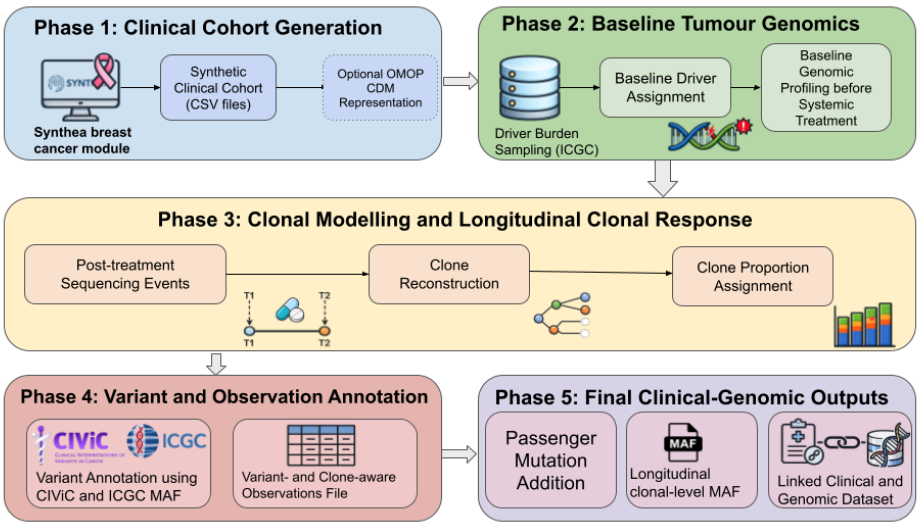

Figure 1: Overview of the full workflow used to generate the linked synthetic clinical-genomic breast cancer dataset.

To assign specific driver genes to synthetic patients, two gene pools were created: a non-treatment-specific driver gene pool and treatment-related driver gene pool. For the non-treatment-specific pool, gene-specific probabilities were estimated from recurrent breast cancer driver genes observed in PCAWG/ICGC samples (Zhang *et al.* 2019). The driver mutation file was treated as an event list, where each row represents a driver event in a tumour sample. For each gene, the frequency was calculated as the number of unique breast cancer samples in which that gene appeared at least once. The 26 most frequent breast cancer driver genes were then used as the non-treatment-specific driver pool, excluding non-gene driver elements such as cytobands or telomeric regions. These probabilities were used to sample background driver genes for each synthetic tumour.

A separate treatment-related gene set was created for therapy categories represented in the Synthea breast cancer module, including chemotherapy, endocrine therapy, HER2-targeted therapies and CDK4/6 inhibitors. Treatment groups were used instead of individual therapies because, at diagnosis where baseline driver genes are assigned, the patient's exact future medications are not yet known. However, the likely therapy category can be inferred from clinical variables already generated by Synthea, such as cancer stage, hormone receptor status, HER2 status and triple-negative status. These same variables were used to determine each patient's eligibility for each treatment group. The specific genes in these groups were selected from CIViC (Griffith *et al.* 2017) and supporting literature based on reported sensitivity or resistance associations with the therapies represented in Synthea. Only gene-disruptive or coding small-variant mechanisms were considered, and associations based on overexpression or underexpression were not included. The treatment-related genes can overlap between groups and with the non-treatment-specific gene pool, since some genes can be relevant to more than one therapeutic context. However, no gene duplicates are allowed for the same patient.

The treatment-related gene groups used by the driver-assignment module are summarized in Table 1. Genes were grouped by Synthea therapy category and by their use in the simulator as sensitivity- or resistance-associated candidates. The draw columns define the order in which treatment-related genes are sampled within each eligible therapy group. Genes supported by sensitivity or resistance evidence for more therapies within the same group were prioritized in the first draw. The two-draw structure was implemented to increase patient-level diversity by preventing every eligible patient from receiving the same full treatment-related gene set.

2.3 Tumour driver assignment at diagnosis

The standard Synthea breast cancer pathway was extended using two custom JSON submodules: “tumor_driver_assignment.json” and “tumor_response_sequencing.json”.

The driver-assignment module is called after diagnosis, staging, hormone receptor characterization and assignment of the internal treatment-response trajectory. The module records a tumour genomic profiling procedure at the time of diagnostic biopsy, before any systemic treatment. This represents the baseline tumour genomic profile for the simulator. The module samples the total number of driver genes from the predefined empirical distribution for each patient and divides the driver set into treatment-related and non-treatment-specific drivers. Treatment-related genes are assigned first and are selected from the defined therapy-specific groups according to the patient’s simulated clinical phenotype. Patients with an “Improving” treatment trajectory receive sensitivity-associated treatment genes, while patients with a “Worsening” trajectory receive resistance-associated genes. The treatment-response trajectory, and therefore the selection of treatment-related genes, is driven by the internal Synthea variable “breast_cancer_survival”, where “yes” corresponds to an improving trajectory and “no” to a worsening trajectory.

Treatment Group	Clinical Eligibility	Therapies	Sensitivity - Draw 1	Sensitivity - Draw 2	Resistance - Draw 1	Resistance - Draw 2
Chemotherapy Group	Triple Negative (HER2-, ER-, PR-) or Stage III or Stage IV	Doxorubicin Epirubicin Paclitaxel	TP53	-	TP53 CHEK2	-
Hormone Therapy	Estrogen Receptor (ER) + or Progesterone Receptor (PR) +	Tamoxifen Letrozole Anastrozole Exemestane Fulvestrant	ESR1 GATA3 MAP3K1 MAP2K4	PIK3CA STK11 AKT1	ARID1A RUNX1 CTCF	TP53 NF1 ESR1
HER2+ Therapies	HER2+	Trastuzumab Ado-trastuzumab Lapatinib Neratinib	ERBB3 PIK3CA	FCGRCA	PTEN PIK3CA ERBB2	-

CDK4/6 Inhibitors	ER+ and PR+	Palbociclib Ribociclib Verzenio	PIK3CA ESR1	-	RB1 FAT1 PTEN NF1	-

Table 1. Treatment-related gene pools used for therapy-aware driver assignment

To avoid making the tumour profile dominated by treatment-related genes, their number is controlled using a predefined budget derived from the sampled total driver count. The module maps each total driver count to a corresponding maximum number of treatment-related assignments, so patients with fewer total drivers also receive a smaller treatment-related driver budget.

The module first goes through each eligible treatment group and assigns one gene per group where possible. If the treatment-related gene budget has not been reached, it then goes through the eligible groups again and can assign a second gene per group. No more than two treatment-related genes can be assigned for a single eligible therapy group. This sequential assignment strategy allows treatment-associated genomic features to be represented while preserving a substantial contribution from non-treatment-specific recurrent breast cancer drivers. Once the treatment-related budget is filled, or no further treatment-related genes can be assigned, the remaining driver slots are filled from the weighted non-treatment-specific breast cancer driver pool. Duplicate gene assignment was prevented using per-gene Boolean attributes. The final gene set was stored as patient attributes and exported in “observations.csv”.

2.4 Longitudinal sequencing and clonal trend assignment

The second custom module, “tumor_response_sequencing.json”, is used to simulate longitudinal tumour genomic behaviour after treatment. This module does not assign new genes. Instead, it reads the driver genes already assigned to the patient at diagnosis and classifies their behaviour after treatment exposure.

Two post-diagnosis sequencing events were added. A second sequencing event can occur for patients treated with neoadjuvant chemotherapy and is implemented as a post-chemotherapy, post-surgical sequencing assessment. A third sequencing event can occur during later post-treatment surveillance after additional systemic therapies, such as endocrine therapy, HER2-targeted therapy or CDK4/6 inhibitor treatment.

For each assigned gene, the module emits a post-treatment clonal trend observation. The possible trend states are “decreasing”, “increasing”, “stable” and “unknown”. A “decreasing” trend represents a treatment-sensitive clone expected to contract after therapy. An “increasing” trend represents a treatment-resistant clone expected to expand. “Stable” is used when no major directional change is expected, and when a treatment-related gene is relevant to the therapy context but does not yet have an active “increasing” or “decreasing” clone trend at this timepoint because the treatment has not been applied yet. “Unknown” is used when a gene has no active treatment-specific interpretation at that timepoint and mainly captures non-treatment-specific drivers or otherwise non-informative genes.

The clonal trend assignment depends on treatment exposure, sequencing phase, and simulated response trajectory. In this way, the genomic layer is generated conditionally on the clinical pathway rather than sampled independently.

2.5 Clone reconstruction and clonal proportion assignment

After Synthea generation, a post-processing pipeline is used to convert gene-level outputs into a clone-aware genomic dataset. The pipeline uses Synthea’s “observations.csv” file as input, which contains both the genomic sequencing observations at diagnosis produced by the driver-assignment module and the later sequencing observations recording post-treatment clonal trends. The script “build_breast_cancer_clones.py” then reconstructs patient-specific clone groups from the genomic observations obtained at diagnosis and post-treatment clonal trend observations.

The clone reconstruction follows a simplified tumour evolution model. A founding clone is created first and is intended to contain early truncal events. Later clones are then created from genes with shared temporal behaviour. Genes with informative longitudinal behaviour are primarily grouped into the same clone when they share identical clonal trajectories across available sequencing timepoints, while some non-informative genes, with an “unknown” clonal trends, may be merged heuristically into existing founding or late clones when clone capacity allows.

The founding clone preferentially includes recurrent and biologically plausible early breast cancer drivers, based on a literature-informed founder-gene prioritisation. Where possible, the founding clone is composed of non-treatment-specific drivers, with a maximum of three genes. Treatment-related genes are generally placed into later clones, since their trends suggest sensitivity or resistance to therapy. Remaining non-treatment-specific genes are added to existing clones, subject to clone-size constraints. Patients with only one or two driver genes are kept as a single clone, while

unusually large clones were split to avoid unrealistic clone structures. In the current implementation, the maximum allowed number of genes per later clone is five.

Clone proportions are assigned using “build_breast_cancer_clone_proportions.py”. For each patient and sequencing timepoint, the script assigns a relative abundance to each clone and normalizes the values so that clone proportions sum to 100%. The temporal behaviour of each clone is constrained by its trend state: increasing clones are initialized at lower abundance and expand over time, while decreasing clones start at higher abundance and contract. Stable clones are constrained to remain around their baseline proportion rather than being allowed to drift freely. Additional rule-based adjustments are then applied to maintain biologically plausible trajectories, including limiting midpoint and final abundances for decreasing clones and preferentially using the founding clone as a buffer when proportions must be redistributed to preserve a total of 100%. These clone-level proportions are used as gene-level clonal abundance proxies, so genes within the same clone share the same longitudinal abundance pattern.

2.6 Variant annotation and pruned observations

The custom Synthea module assigns driver genes, but not exact genomic variants. Therefore, after each Synthea run, every assigned driver gene is mapped to a specific variant using variant reference libraries. The first library, “breast_cancer_driver_variants_from_maf.csv”, was used as the general driver variant library. The second library, “civic_breast_cancer_driver_variants_from_maf.csv”, was used as the therapy-specific variant library for treatment-linked genes when a sensitivity or resistance interpretation was available.

The general driver variant library was built from ICGC breast cancer MAF data and the ICGC panorama of driver mutations reference table. Where possible, known driver mutations from the reference table were matched back to exact MAF rows. When no exact driver entry was available, coding or gene-disruptive variants were extracted from the MAF for that gene, including missense mutations, nonsense mutations, frameshift variants, splice-site variants, in-frame insertions or deletions and related coding-impact classes. These classes were selected because they are more likely to alter protein function directly than synonymous or non-coding variants. Breast cancer variants were prioritized. When no breast cancer variant was available for a gene, mainly variants from other carcinoma datasets were used as fallback.

The therapy-specific variant library was built from a manually curated set of CIViC molecular profiles associated with sensitivity or resistance to therapies represented in the Synthea breast cancer

pathway. Not all therapies had corresponding CIViC-derived variants, either because no relevant sensitivity or resistance variant was identified in CIViC or because the CIViC variant could not be matched to the local ICGC MAF. CIViC variants were matched to the MAF using exact allele information, including chromosome, start position, end position, reference allele and alternate allele. CIViC profiles that could not be matched to an exact MAF row were excluded from automatic assignment.

Exact variant assignment is performed after the Synthea run. For each treatment-linked gene, the pipeline first attempts to select a compatible variant from the CIViC-derived therapy-specific library. The choice depends on the patient's medication history and latest directional clonal trend: decreasing trends are interpreted as treatment sensitivity, while increasing trends are interpreted as resistance. Only medications started on or before the date of that latest directional trend are considered when matching a CIViC-compatible variant. If no suitable CIViC-compatible variant is available for that gene and treatment context, the pipeline selects a variant from the general driver variant library instead. Non-treatment-specific driver genes are assigned variants directly from the general driver variant library and do not trigger CIViC matching. CIViC-compatible variants are selected randomly among eligible therapy-specific matches, whereas fallback selection from the general driver variant library is deterministic for each patient-gene combination.

The script "build_breast_cancer_pruned_observations.py" merges the original Synthea "observations.csv" file with clone proportions and variant annotations to create the "observations_pruned_by_clone.csv" for each run. Non-genomic observations are kept unchanged. Only the genomic sequencing content is updated and rewritten into a more standardized clinical-data format. During this step, a 5% clone proportion threshold is applied at each sequencing timepoint. Genomic observations below this threshold are removed from the final clinical output so that low-abundance variants are not displayed as detected events. This filtering is timepoint-specific: variants belonging to increasing clones can be absent from earlier sequencing observations but appear at later timepoints once their clone proportion exceeded 5%, while variants belonging to decreasing clones can be visible at earlier timepoints and disappear later as their clone proportion fell below the threshold. The resulting file rewrites the genomic sequencing content into variant-oriented observations containing gene name, genomic change, variant classification, clonal role and estimated clonal prevalence.

2.7 Building the final patient Mutation Annotation Format (MAF) files

Passenger mutations were added separately. A passenger-only breast cancer mutation pool was created from the ICGC MAF by retaining only breast cancer samples and removing all rows whose genes appeared as drivers in the ICGC driver reference table. Each synthetic patient with genomic sequencing was assigned deterministically to one source breast tumour sample from this passenger pool, and that sample's complete passenger mutation set was transferred to the synthetic patient. This approach was used to preserve a realistic passenger mutation background without independently sampling an arbitrary number of unrelated passenger variants.

To create the final patient-level MAF outputs, the assigned passenger mutation set was then combined with the driver variants retained in "observations_pruned_by_clone.csv" and the assigned clone groups. One MAF file is written per reconstructed clone for each patient. Driver variants are assigned to clone-specific MAF files according to the clone group of their genes.

Driver variants are reconstructed by matching each retained event-specific observation back to an exact source row from the ICGC-derived variant libraries using the combination of gene symbol, genomic DNA change, reported using genomic Human Genome Variation Society notation (gHGVS) and variant classification. Matching is attempted against the curated general driver variant library and the CIViC-derived therapy-specific library, with an exact match to the original raw MAF used as a fallback if a retained driver could not be recovered from the curated source tables.

Passenger mutations are split randomly but without replacement across the patient's reconstructed clones, so each passenger mutation appears in only one clone-level MAF file. The output therefore consists of per-patient, per-clone MAF files in which each file represents the driver and passenger mutations assigned to one reconstructed tumour clone.

The pipeline is stochastic, so each execution can produce a different synthetic cohort. This variability comes from the underlying Synthea simulation, which generates different synthetic patients on each run, and from the added tumour driver assignment module, which samples driver genes from treatment-related and non-treatment-specific gene pools. When therapy-specific CIViC matches are available, one eligible CIViC variant is selected at random for the corresponding patient-gene combination. Otherwise, variants are assigned deterministically from the general driver variant library for each patient-gene combination. As a result, the exact cohort composition, assigned genes, variants and clonal structures may differ between runs. Example cohorts are provided in the project repository, and users can also run the simulator themselves by following the instructions in the repository README.md.

2.8 Loading Synthea data into OMOP CDM

The generated Synthea CSV files can be transformed into the Observational Medical Outcomes Partnership Common Data Model (OMOP CDM) (Voss et al. 2015). The OMOP CDM is used to provide a standardized structure for clinical concepts and to support compatibility with observational health data research tools.

Synthea version 3.3.0 was selected because it's supported by ETL-Synthea, an OHDSI tool for converting Synthea output to OMOP CDM format. The Synthea CSV files were transformed using the OHDSI ETL-Synthea package. The tool supports loading the data into OMOP CDM version 5.3 and 5.4 (Molinaro et al. 2024). In this project, the native Synthea CSV files were read by ETL-Synthea, mapped to OMOP vocabularies and loaded into a PostgreSQL database. Athena vocabularies were downloaded for the tool to work and a local Postgres database had to be created. After the tool ran, the database was populated with the original Synthea-derived clinical data together with the corresponding OMOP-mapped representation. Genomic events generated in this project were designed to be compatible with the OMOP CDM structure and were represented in the "MEASUREMENT" table.